ECE 720 Advanced Topics in Software Engineering and Intelligent Systems

Fall 2026 - September 1 to December 8

Paper review (4% each, 20% in total)

Description

Given that this is a seminar-style course, you are expected to write **five** paper reviews throughout the semester. Before coming to class, read each paper to the best of your ability. You are not expected to understand every detail, but you should be able to convey your understanding of the paper's core **motivations** and **innovations**. The goal of paper review is to help you articulate your opinion of the paper in order to spark in-class discussion.

Remember, the **in-class discussion** is meant to simulate the *reviewer discussion* from the typical review process for software engineering conference papers. You should put yourself in the shoes of a reviewer. Be critical and provide constructive feedback. To guide you through this process, make sure to structure your paper response as follows:

- *Summary (of the paper)*
- *Strengths*
- *Weaknesses*
- *Comments* (Refer to Review Criteria from ICSE 2027)
    - *Novelty*
    - *Rigor*
    - *Relevance*
    - *Verifiability and Transparency*
    - *Presentation*

Evaluation scheme (4pt)

You will be evaluated on the following criteria:

- *Summary of the paper (1 pt)*
    - Clearly and accurately summarize the paper's problem, motivation, approach, and key contributions.
    - Demonstrate your own understanding of the paper rather than copying or closely paraphrasing the abstract.
    - Focus on the main ideas rather than implementation details.
- *Strengths (0.5 pt)*
    - Identify at least three specific strengths of the paper.
    - Briefly explain **why each strength matters** to the quality or contribution of the work.
    - Avoid generic statements such as "the paper is well written" without explanation.
- *Weaknesses (0.5 pt)*
    - Identify at least three specific weaknesses, limitations, or concerns.

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  - Briefly explain **why each issue matters** and how it affects the paper's claims, validity, or impact.
  - Distinguish substantive concerns from minor presentation issues.
- *Comments (2 pt): Provide a critical and constructive evaluation of the paper. Your comments should explicitly consider the ICSE 2027 review criteria.*
  - Novelty: Is the work sufficiently original compared with prior research?
  - Rigor: Are the methodology, technical approach, and evaluation sound and convincing?
  - Relevance: Is the problem important, and does the work have meaningful implications for software engineering?
  - Verifiability & Transparency: Does the paper provide sufficient information and evidence to support and independently examine its claims?
  - Presentation: Is the paper clearly organized, explained, and argued?
  - Do not simply state whether the paper performs well or poorly on these criteria. Explain your reasoning, connect your comments to specific parts of the paper, and suggest concrete improvements where appropriate.

Tips for a strong paper review

Be **specific** in your response, and remember the **examples of bad reviews and reviewers** we discussed in class. Additional advice for reading and critiquing papers:

- How to read a paper
- The task of the referee

Submission

Paper reviews must be submitted to Canvas ***a week before the lecture*** in which the paper is discussed. Specific deadlines are listed on Canvas under:

- Assignments → Paper Reviews → Paper [#] Due [Date]